

Blur Detection Batch Processing System

Non-Technical / Quality Inspector User Manual
(Position & Distance Definition Revised Edition)

Welcome to the **Blur Detection Batch Processing System**. This system integrates advanced Artificial Intelligence algorithms (Meta SAM3 and DINOv3) to automatically identify targets within photos, detect whether the image exhibits offset or distance anomalies, and automatically categorize them. This manual is written specifically for non-technical operators and covers system startup, as well as subsequent manual data review and calibration guidelines.

I. Core Precautions (Must Read First)

1. Image Naming Convention (Crucial):

All pending image filenames must strictly follow the **SerialNumber-Position** format (the serial number and position must be separated by a hyphen -).

- **Correct Examples:** **1359-3.jpg** (Serial number is 1359, position is 3), **4582A-9.bmp**
- **Incorrect Examples:** **image_3.jpg** , **1359_pos3.png**
- **If this naming convention is not met, the system will fail to correctly parse the position, resulting in processing failure!**

2. Black Command Line Window:

WARNING: Throughout the entire workflow, a black Command Prompt (Terminal) window will pop up. **Do not close it before the work is finished!** Closing this window will directly and forcibly terminate the background AI analysis program.

II. Data Operation Steps

1. Start the Program

In the program's root directory, locate the file named **blur_detection.bat** and double-click to run it.

2. Select Drive Type

The black window will prompt: *"Is your data drive external or internal?"*

- If the photos are on an external hard drive / USB flash drive / portable SSD, please press the letter **E** on your keyboard.
- If the photos are on the computer's local disk (e.g., C drive, D drive), please press the letter **I** (or **1**) on your keyboard.

3. Select Input Folder and Run

In the folder selection window that pops up, select the folder containing the photos to be inspected and click **OK**. Wait 2-3 seconds, and your browser will automatically open the management dashboard (the address is usually <http://localhost:8501>). Click the **"Run Batch"** button on the webpage and wait for the progress bar to complete.

III. Image Review & Annotation Guidance

After the batch processing is completed, the operator needs to switch to the **Image Review & Annotation** panel to manually approve or correct the AI's prediction results.

Panel Operations:

- **Image Zoom:** Hover your mouse over the local slice image of each position, and the image will automatically zoom to full screen. No clicking is necessary; moving the mouse away will restore it.
- **Manual Calibration Labels:** If you find that the AI's prediction is inaccurate, directly click the corresponding **0**, **F**, **SN**, or **N** buttons on the left side of the picture. Once clicked, the label for that position will be instantly overridden and corrected.
- **Save & Turn Page:** Click the **Next** button (or directly press the keyboard  arrow key), and the system will automatically summarize and save all annotations for the current product and seamlessly switch to the next order. Click **Previous** (or the keyboard  key) to return to the previous record.

1. Assessment Criteria

When applying manual calibration labels, please strictly adhere to the following four quality inspection position criteria:

Manual Label	Class (Status)	Visual Judgment & Position/Distance Feature Description
0	OK (Normal)	The target position is perfectly standard, with no offset or distance anomaly whatsoever.
F	Far (Too Far)	The target has severely offset or the distance is too far, causing feature misalignment or a missing target area.
SN	Slightly Near	The target distance is slightly near. The boundary outline is still recognizable, but it is approaching the threshold edge.
N	Near (Too Near)	The target distance is severely too near, resulting in extreme defocus, feature stacking, magnification, or loss of the core field of view.

2. Auto-FLC Auto-Linkage Rules

The system has a rigorous built-in algorithm linkage mechanism. Across the five grid views (1, 3, 5, 7, 9), if the following conditions are triggered, the system will automatically force the unit to be locked into the **Need FLC** (Requires Anomaly Review) status:

- Any **1** position is manually labeled/modified to **N (Near)** by the operator.
- A cumulative of **3 or more** positions are manually labeled/modified to **SN (Slightly Near)** by the operator.

***Note:** If none of the above conditions are triggered, and the operator does not forcefully override the results, the unit will be automatically and safely cleared upon turning the page and marked as **Pass (Human Reviewed)**.*

IV. View Quality Inspection Report Results

After the review concludes, the operator can check the following files in the `pipeline_outputs/dataset_output/` directory:

- `human_annotated_predictions.csv` : **Final Human Inspection Roster.** It completely records the final status of every single product after manual correction and confirmation.
- `consolidated_batch_predictions.csv / .xlsx` : Contains the raw AI detection results, confidence scores, and processing suggestions for all images.